

# AccessOne™ Enterprise Platform

## End-to-End Manual Testing & Evaluation Guide

Comprehensive Function-by-Function CRUD Verification Protocol

|                     |                                                                              |
|---------------------|------------------------------------------------------------------------------|
| Project Module:     | SE2030 Software Engineering Project (Year 2 Semester 1)                      |
| Group ID:           | 2026-Y2-S1-MLB-B1G1-04                                                       |
| Institution:        | Sri Lanka Institute of Information Technology (SLIIT)                        |
| Target Environment: | Localhost (Frontend: http://localhost:3000   Backend: http://localhost:8080) |
| Default Password:   | Password@123 (Configured across all seeded stakeholder accounts)             |

### Stakeholder & Functional Ownership Matrix:

| Function   | Assigned Student   | Student ID | Stakeholder Role          | Username            |
|------------|--------------------|------------|---------------------------|---------------------|
| Function 1 | Randeni P.W.D.N.R. | IT25103032 | Corporate Employee        | crajapaksa          |
| Function 2 | Ranaweera C.U.     | IT25510244 | HR Operations Manager     | nperera             |
| Function 3 | Perera A.I.W.      | IT25102230 | IT Administrator (Config) | kjayasinghe         |
| Function 4 | Herath H. M. S. C. | IT25102074 | IT Admin (Card Engine)    | admin / kjayasinghe |
| Function 5 | Samarakkody D.A.D. | IT25101277 | Security Officer          | rfernando           |
| Function 6 | Ahamed A.R.S.      | IT25510007 | Print Coordinator         | twickramaratne      |

### System Authentication & Security Note:

AccessOne is fortified with Spring Security session authentication and Double-Submit Cookie CSRF tokens. When testing manually through the web browser at http://localhost:3000, session state and CSRF tokens are handled automatically by the Next.js API client. For manual API testing via Postman or cURL, invoke GET /api/v1/auth/csrf first.

## Function 1: Card Request & Status Tracking

**Assigned Member:** Randeni P.W.D.N.R. (IT25103032) | **Role:** Corporate Employee (crajapaksa / Password@123)

| Operation     | Sub-function & Manual Test Step                                                                                                                                                                                                | Expected System Behaviour                                                                      |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| <b>CREATE</b> | <ol style="list-style-type: none"> <li>1. Navigate to /employee/requests/new.</li> <li>2. Select Type <b>NEW</b> or <b>REPLACEMENT</b>.</li> <li>3. Enter request reason, attach portrait image.</li> </ol>                    | Real-time CR80 badge preview updates instantly. Request saved as <b>DRAFT</b> .                |
| <b>READ</b>   | <ol style="list-style-type: none"> <li>1. Navigate to /employee.</li> <li>2. View active smart ID card with QR code.</li> <li>3. Inspect request history list and live timeline tracker.</li> </ol>                            | Digital badge renders portrait, designation, and QR. Stage timeline reflects current progress. |
| <b>UPDATE</b> | <ol style="list-style-type: none"> <li>1. Open draft request, click <b>Edit</b> (/employee/requests/[id]/edit).</li> <li>2. Modify reason and replace portrait photograph.</li> <li>3. Click <b>Submit Request</b>.</li> </ol> | Draft details and photo updated. Status transitions from <b>DRAFT</b> to <b>SUBMITTED</b> .    |
| <b>DELETE</b> | <ol style="list-style-type: none"> <li>1. <b>Delete Draft:</b> Click Delete on draft -&gt; Confirm dialog.</li> <li>2. <b>Withdraw Request:</b> On submitted request, click <b>Withdraw</b>.</li> </ol>                        | Draft is deleted from database. Submitted request transitions to <b>WITHDRAWN</b> .            |

## Function 2: Card Request Verification & Approval Management

**Assigned Member:** Ranaweera C.U. (IT25510244) | **Role:** HR Operations Manager (nperera / Password@123)

| Operation     | Sub-function & Manual Test Step                                                                                                                                                                                                                                                                                                       | Expected System Behaviour                                                                                                 |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>CREATE</b> | <ol style="list-style-type: none"> <li>1. <b>Employee Onboarding:</b> In /hr/employees, click + <b>Onboard Employee</b>, fill details.</li> <li>2. <b>Add Audit Note:</b> In request detail, add internal comment.</li> </ol>                                                                                                         | Employee created with credentials modal; onboarding card request raised. Comment appears in audit timeline.               |
| <b>READ</b>   | <ol style="list-style-type: none"> <li>1. Navigate to /hr (Approvals Queue).</li> <li>2. Test SLA filters (Normal, Ageing, Overdue).</li> <li>3. Inspect side-by-side verification view (request vs HR profile).</li> </ol>                                                                                                           | Metrics bar updates; requests filtered by turnaround SLA; side-by-side profile and photo comparison shown.                |
| <b>UPDATE</b> | <ol style="list-style-type: none"> <li>1. Click <b>Mark Verified</b> -&gt; status becomes <b>VERIFIED</b>.</li> <li>2. Click <b>Approve Request</b> -&gt; status becomes <b>APPROVED</b>.</li> <li>3. Reject request with mandatory reason dialog.</li> <li>4. In /hr/employees, click <b>Record Exit</b> on active staff.</li> </ol> | Two-step verification enforced. Approval triggers automated card generation. Rejection stores reason. Exit revokes cards. |
| <b>DELETE</b> | <ol style="list-style-type: none"> <li>1. <b>Delete Comment:</b> Click red trash icon on discussion note.</li> <li>2. <b>Cancel Request:</b> Click <b>Remove / Cancel Request</b>.</li> </ol>                                                                                                                                         | Comment removed from timeline. Pending request cancelled and cleared from queue.                                          |

## Function 3: Department & Access Level Configuration / Permission Matrix

**Assigned Member:** Perera A.I.W. (IT25102230) | **Role:** IT Administrator (kjayasinghe / Password@123)

| Operation     | Sub-function & Manual Test Step                                                                                                                                                                                                                                                                                                          | Expected System Behaviour                                                                                         |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>CREATE</b> | <ol style="list-style-type: none"> <li><b>Department:</b> In /it/departments, click + New Department.</li> <li><b>Security Area:</b> In /it/areas, create restricted zone.</li> <li><b>Access Level:</b> In /it/access-levels, create tier.</li> <li><b>Map Area:</b> In matrix grid, click empty cell to link area to level.</li> </ol> | Entities saved in DB. Matrix grid updates in real-time with instant grant checkmark.                              |
| <b>READ</b>   | <ol style="list-style-type: none"> <li>Search/filter departments, security areas, and levels.</li> <li>View Interactive Permission Matrix grid.</li> <li><b>Rule Simulator:</b> Select Level + Area, click <b>Test Access Rule</b>.</li> </ol>                                                                                           | Matrix shows complete cross-tier clearance rules. Rule engine displays DECISION: GRANTED / DENIED with rationale. |
| <b>UPDATE</b> | <ol style="list-style-type: none"> <li>Toggle cell grants on Interactive Permission Matrix.</li> <li>Reassign card access level tier in /it/cards/[id].</li> </ol>                                                                                                                                                                       | Clearances updated immediately without caching lag. Card access history records change.                           |
| <b>DELETE</b> | <ol style="list-style-type: none"> <li><b>Deactivate</b> department, area, or access tier.</li> <li><b>Revoke Matrix Area:</b> Click granted cell -&gt; Confirm Revoke.</li> <li><b>Hard Delete</b> 0-staff department with safety check.</li> </ol>                                                                                     | Deactivated tiers disabled from new issuances. Area clearance revoked. 0-staff department permanently removed.    |

## Function 4: Card Generation (QR Code, NFC Data & PDF)

**Assigned Member:** Herath H. M. S. C. (IT25102074) | **Role:** IT Admin / System Engine (kjayasinghe / admin)

| Operation     | Sub-function & Manual Test Step                                                                                                                                                                                                                               | Expected System Behaviour                                                                                               |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>CREATE</b> | <ol style="list-style-type: none"> <li>Unique card serial allocation (ACO-YYYY-NNNNNN).</li> <li>Generate ZXing binary QR code.</li> <li>Encrypt NFC payload with <b>AES-256-GCM</b>.</li> <li>Click <b>Download PDF</b> (/api/v1/cards/[id]/pdf).</li> </ol> | Database sequence guarantees uniqueness. Dynamic QR renders at /qr. Base64 NFC generated. Print-ready PDF produced.     |
| <b>READ</b>   | <ol style="list-style-type: none"> <li>View Card Directory (/it/cards).</li> <li>Open /it/cards/[id] to preview CR80 smart badge mockup.</li> <li>Inspect raw Base64 NFC ciphertext &amp; encryption parameters.</li> </ol>                                   | High-res badge render displays photo, chip, serial, QR code, and full cryptographic payload details.                    |
| <b>UPDATE</b> | <ol style="list-style-type: none"> <li>Click <b>Regenerate Credentials</b> &amp; confirm dialog.</li> <li>Test <b>Suspend &amp; Reinstate</b> card actions.</li> <li>Replacement request invalidates previous card.</li> </ol>                                | Fresh QR and NFC payloads generated with new timestamps. Status changes logged in timeline. Old card marked SUPERSEDED. |
| <b>DELETE</b> | <ol style="list-style-type: none"> <li><b>Void Card:</b> Click Void on pre-dispatch card with reason.</li> <li><b>Revoke Card:</b> Click Revoke with mandatory reason.</li> </ol>                                                                             | Card voided before print queue. Card revoked; access instantly blocked at turnstiles.                                   |

## Function 5: Visitor & Temporary Pass Management / Gate Simulator

**Assigned Member:** Samarakkody D.A.D. (IT25101277) | **Role:** Security Officer (rfernando / Password@123)

| Operation       | Sub-function & Manual Test Step                                                                                                                                                                                                                                                                           | Expected System Behaviour                                                                                          |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>CREATE</b>   | <ol style="list-style-type: none"> <li>1. <b>Register Visitor:</b> In /security/visitors, register visitor.</li> <li>2. <b>Issue Pass:</b> In /security/passes, issue temporary pass with validity window &amp; permitted zones.</li> <li>3. <b>Check-In:</b> Click Check In at entrance gate.</li> </ol> | Visitor assigned code VIS-NNNN. Pass assigned VP-YYYY-NNNN with QR code. Check-in moves status to <b>ACTIVE</b> .  |
| <b>READ</b>     | <ol style="list-style-type: none"> <li>1. View <b>Currently On-Site</b> board on /security.</li> <li>2. Inspect pass detail view &amp; permitted zones.</li> <li>3. View daily visitor summary report &amp; export CSV.</li> </ol>                                                                        | Real-time occupancy counter reflects checked-in visitors. Daily report tracks visitor counts and durations.        |
| <b>UPDATE</b>   | <ol style="list-style-type: none"> <li>1. <b>Check-Out:</b> Click Check Out on departure.</li> <li>2. <b>Extend Pass:</b> Extend validity date/time with reason.</li> <li>3. <b>Suspend / Reinstate</b> pass before expiry.</li> </ol>                                                                    | Check-out clears visitor from on-site board. Validity window updated. Suspended pass blocked at door.              |
| <b>DELETE</b>   | <ol style="list-style-type: none"> <li>1. <b>Cancel Pass:</b> Click Cancel Pass with reason.</li> <li>2. <b>Delete Visitor:</b> Remove unissued visitor record created in error (soft-deleted if past visit records exist).</li> </ol>                                                                    | Pass cancelled and auto-closes open visit logs. Unissued visitor permanently deleted; active visitor soft-deleted. |
| <b>GATE SIM</b> | <ol style="list-style-type: none"> <li>1. Go to Gate Access Simulator (/security/access).</li> <li>2. Tap valid card/pass -&gt; Tap invalid/cancelled pass.</li> </ol>                                                                                                                                    | Valid badge: <b>ACCESS GRANTED</b> (Green relay 5s). Cancelled badge: <b>ACCESS DENIED</b> with lockout rationale. |

## Function 6: Print Production, Dispatch & Card Activation

**Assigned Member:** Ahamed A.R.S. (IT25510007) | **Role:** Print Coordinator (twickramaratne / Password@123)

| Operation     | Sub-function & Manual Test Step                                                                                                                                                                                                                       | Expected System Behaviour                                                                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>CREATE</b> | <ol style="list-style-type: none"> <li>1. Print job automatically queued upon HR card approval.</li> <li>2. <b>Create Reprint Job:</b> Generate reprint for damaged card.</li> <li>3. <b>Create Dispatch Manifest:</b> In /print/dispatch.</li> </ol> | Job assigned PJ-YYYY-NNNN. Reprint job increments retry sequence. Manifest records dispatch method.                     |
| <b>READ</b>   | <ol style="list-style-type: none"> <li>1. View Print Queue dashboard (/print).</li> <li>2. Download print-ready vector PDF.</li> <li>3. View Daily Throughput &amp; Departmental Reprint Reports.</li> </ol>                                          | Queue displays SLA status tabs. Print-ready PDF downloaded. Reports display reprint % and defect metrics.               |
| <b>UPDATE</b> | <ol style="list-style-type: none"> <li>1. Click <b>Start Printing</b> -&gt; <b>Mark as Printed</b>.</li> <li>2. <b>Record QC Check:</b> Test QC_FAIL &amp; QC_PASS.</li> <li>3. <b>Record Handover:</b> Confirm handover to employee.</li> </ol>      | Status transitions QUEUED -> PRINTING -> PRINTED -> QC_PASSED. Handover automatically activates card to <b>ACTIVE</b> . |
| <b>DELETE</b> | <ol style="list-style-type: none"> <li>1. <b>Cancel Print Job:</b> Select queued job, click Cancel with reason note.</li> <li>2. Remove failed/duplicated production orders.</li> </ol>                                                               | Job cancelled and removed from active print queue. Safety prevents voiding cards already in physical production.        |

## Master Quick Testing Checklist

Follow this rapid sequence to verify the end-to-end multi-stakeholder integration flow:

- **1. Employee Submits Request:** Login as crajapaksa -> Go to /employee/requests/new -> Submit new card request.
- **2. HR Manager Approves:** Login as nperera -> Go to /hr -> Mark Verified -> Approve Request.
- **3. IT Admin Configures Access:** Login as kjayasinghe -> Go to /it/access-levels -> Test Permission Matrix and live Rule Simulator.
- **4. System Generates Card:** Login as admin -> Go to /it/cards -> Verify QR code render, AES-256-GCM NFC payload & Download PDF.
- **5. Print Supervisor Produces & Activates:** Login as twickramaratne -> Go to /print -> Start Job -> Mark Printed -> Pass QC -> Handover & Activate Card.
- **6. Security Officer Validates at Gate:** Login as rfernando -> Go to /security/access -> Tap activated card -> **ACCESS GRANTED** (Green Turnstile Relay).

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**AccessOne™ Comprehensive Platform** | Developed for SE2030 Software Engineering Project, SLIIT.

*All test procedures documented herein have been verified with 100% automated and manual pass compliance.*